Problem 1. ***Matching Estimators.***

Consider the usual setup from causal inference: you observe (Y_i, W_i, X_i) for an independent sample of n units, where $W_i = 1$ if unit i was treated and 0 otherwise, $Y_i = W_iY_i(1) + (1 - W_i)Y_i(0)$ is the realized outcome for individual i, where $Y_i(1)$ and $Y_i(0)$ are individual i's treated and untreated potential outcomes, respectively, and X_i is a vector of controls that satisfy unconfoundedness: $(Y_i(1), Y_i(0))$ and W_i are independent conditional on X_i. The goal is to estimate the average treatment effect on the treated

$$\tau = E[Y_i(1) - Y_i(0)|W_i = 1].$$

The implicit assumption is that $E[|Y_i(0)|], E[|Y_i(1)|] < \infty$ so that τ is well defined. A common empirical strategy is to use a matching estimator:

$$\widehat{\tau} = \frac{1}{n_1}\sum_{i=1}^{n} W_i \left(Y_i - \frac{1}{M}\sum_{j\in N(i)} Y_j \right),$$

where $n_1 = \sum_{i=1}^{n} W_i$ are the number of treated units in the sample, $M \geq 1$ are the number of units to match on, and $N(i)$ indexes the M untreated units closest to individual i, where "distance" is measured by covariates. That is, each $j \in N(i)$ has $W_j = 0$ and $\|X_i - X_j\| \leq \|X_i - X_k\|$ for any untreated unit $k \notin N(i)$. This averaging over a data-dependent neighborhood introduces a lot of dependence across observations in the sum, making matching estimators difficult to analyze.

The following steps work through a martingale construction from Abadie and Imbens (2012). This may be used to derive the asymptotic distribution of $\widehat{\tau}$. Let $\mu_d(X_i) = E[Y_i(d)|X_i]$. By addition and subtraction of terms, the decomposition is

$$\sqrt{n_1}(\widehat{\tau} - \tau) = \frac{1}{\sqrt{n_1}}\sum_{i=1}^{n} W_i(\mu_1(X_i) - \mu_0(X_i) - \tau) + \frac{1}{\sqrt{n_1}}\sum_{i=1}^{n} W_i \left(Y_i - \mu_1(X_i) - \frac{1}{M}\sum_{j\in N(i)} (Y_j - \mu_0(X_j)) \right) + \textit{remainder},$$

where

$$\textit{remainder} = \frac{1}{\sqrt{n_1}}\sum_{i=1}^{n} W_i \left(\mu_0(X_i) - \frac{1}{M}\sum_{j\in N(i)} \mu_0(X_j) \right)$$

is a bias term that comes from interpolating the conditional mean of unit i with the conditional means of its neighbors in $N(i)$.

(a) Show that the second sum above can be rewritten in terms of $K_{n,i} = \sum_{j=1}^{n} I[i \in N(j)]$ to give

$$\sqrt{n_1}(\widehat{\tau} - \tau) = \frac{1}{\sqrt{n_1}}\sum_{i=1}^{n} W_i(\mu_1(X_i) - \mu_0(X_i) - \tau) + \frac{1}{\sqrt{n_1}}\sum_{i=1}^{n} \left(W_i - (1 - W_i)\frac{K_{n,i}}{M} \right)(Y_i - \mu_{W_i}(X_i)) + \textit{remainder}.$$

Hint: this is just algebra. Nothing deep.

(b) *Define* $X_{i,n} = \sum_{j=1}^{i} \varepsilon_{j,n}$ *for* $i = 1, \dots, 2n$, *where*

$$\varepsilon_{i,n} = \begin{cases} \frac{1}{\sqrt{n_1}} W_i(\mu_1(X_i) - \mu_0(X_i) - \tau), & 1 \le i \le n, \\ \frac{1}{\sqrt{n_1}} \left(W_{i-n} - (1 - W_{i-n}) \frac{K_{n,i-n}}{M} \right) (Y_{i-n} - \mu_{W_{i-n}}(X_{i-n})), & n+1 \le i \le 2n. \end{cases}$$

Also define

$$\mathcal{F}_{i,n} = \begin{cases} \sigma(W_1, \dots, W_n, X_1, \dots, X_i), & 1 \le i \le n, \\ \sigma(W_1, \dots, W_n, X_1, \dots, X_n, Y_1, \dots, Y_{i-n}), & n+1 \le i \le 2n. \end{cases}$$

Show that $\{(X_{i,n}, \mathcal{F}_{i,n}) : 1 \le i \le 2n\}$ *is a martingale.*

Solution.

(a) Begin with the second sum:

$$\frac{1}{\sqrt{n_1}} \sum_{i=1}^{n} W_i \left(Y_i - \mu_1(X_i) - \frac{1}{M} \sum_{j \in N(i)} (Y_j - \mu_0(X_j)) \right).$$

Splitting this into two parts yields:

$$= \frac{1}{\sqrt{n_1}} \sum_{i=1}^{n} W_i(Y_i - \mu_1(X_i)) - \frac{1}{\sqrt{n_1}} \sum_{i=1}^{n} W_i \frac{1}{M} \sum_{j \in N(i)} (Y_j - \mu_0(X_j)).$$

The second term involves summing over treated units i and their untreated matches j. Observe that each untreated unit j appears as a match exactly $K_{n,j}$ times across all treated units. Therefore, the sum can be reindexed by switching the order of summation:

$$\sum_{i=1}^{n} W_i \frac{1}{M} \sum_{j \in N(i)} (Y_j - \mu_0(X_j)) = \sum_{j=1}^{n} (1 - W_j) \frac{K_{n,j}}{M} (Y_j - \mu_0(X_j)).$$

Combining the two parts:

$$\begin{aligned} &\frac{1}{\sqrt{n_1}} \sum_{i=1}^{n} W_i(Y_i - \mu_1(X_i)) - \frac{1}{\sqrt{n_1}} \sum_{j=1}^{n} (1 - W_j) \frac{K_{n,j}}{M} (Y_j - \mu_0(X_j)) \\ &= \frac{1}{\sqrt{n_1}} \sum_{i=1}^{n} \left[W_i(Y_i - \mu_1(X_i)) - (1 - W_i) \frac{K_{n,i}}{M} (Y_i - \mu_0(X_i)) \right] \\ &= \frac{1}{\sqrt{n_1}} \sum_{i=1}^{n} \left[W_i - (1 - W_i) \frac{K_{n,i}}{M} \right] (Y_i - \mu_{W_i}(X_i)), \end{aligned}$$

where the last equality uses the fact that when $W_i = 1$, the term becomes $W_i(Y_i - \mu_1(X_i))$, and when $W_i = 0$, it becomes $-(1 - W_i) \frac{K_{n,i}}{M} (Y_i - \mu_0(X_i))$.

(b) To show that $\{(X_{i,n}, \mathcal{F}_{i,n}) : 1 \leq i \leq 2n\}$ is a martingale, the following properties must be verified:

(i) $X_{i,n}$ is $\mathcal{F}_{i,n}$-measurable and integrable for all i.

(ii) $\mathcal{F}_{i,n} \subseteq \mathcal{F}_{i+1,n}$ for all i.

(iii) $E[X_{i+1,n}|\mathcal{F}_{i,n}] = X_{i,n}$, or equivalently, $E[\varepsilon_{i+1,n}|\mathcal{F}_{i,n}] = 0$.

Properties (i) and (ii) follow directly from the definitions. The focus is on verifying the martingale property (iii).

Case 1: $1 \leq i \leq n-1$

Here $\varepsilon_{i+1,n} = \frac{1}{\sqrt{n_1}} W_{i+1}(\mu_1(X_{i+1}) - \mu_0(X_{i+1}) - \tau)$.

Given $\mathcal{F}_{i,n} = \sigma(W_1, \ldots, W_n, X_1, \ldots, X_i)$, it follows that:

$$E[\varepsilon_{i+1,n}|\mathcal{F}_{i,n}] = \frac{1}{\sqrt{n_1}} W_{i+1} E[\mu_1(X_{i+1}) - \mu_0(X_{i+1}) - \tau | W_1, \ldots, W_n, X_1, \ldots, X_i].$$

Since X_{i+1} is independent of $(X_1, \ldots, X_i)$ (i.i.d. assumption) and W_{i+1} is known (contained in $\mathcal{F}_{i,n}$):

$$E[\mu_1(X_{i+1}) - \mu_0(X_{i+1})|W_{i+1}] = E[\mu_1(X) - \mu_0(X)|W_{i+1}].$$

By definition, $\tau = E[Y_i(1) - Y_i(0)|W_i = 1] = E[\mu_1(X) - \mu_0(X)|W = 1]$.

Under unconfoundedness:

$$E[\mu_1(X) - \mu_0(X)|W = 1] = \tau.$$

Therefore:

$$E[\varepsilon_{i+1,n}|\mathcal{F}_{i,n}] = \frac{1}{\sqrt{n_1}} W_{i+1} \cdot 0 = 0.$$

Case 2: $i = n$

Transitioning from $\mathcal{F}_{n,n} = \sigma(W_1, \ldots, W_n, X_1, \ldots, X_n)$ to $\mathcal{F}_{n+1,n} = \sigma(W_1, \ldots, W_n, X_1, \ldots, X_n, Y_1)$,

$$\varepsilon_{n+1,n} = \frac{1}{\sqrt{n_1}} \left(W_1 - (1 - W_1) \frac{K_{n,1}}{M} \right) (Y_1 - \mu_{W_1}(X_1)).$$

Given $\mathcal{F}_{n,n}$, all W_i, X_i, and $K_{n,i}$ are known. The requirement is:

$$E[Y_1 - \mu_{W_1}(X_1)|\mathcal{F}_{n,n}] = 0.$$

By definition, $\mu_d(x) = E[Y_i(d)|X_i = x]$, and by unconfoundedness:

$$E[Y_1|W_1, X_1] = \mu_{W_1}(X_1).$$

Therefore:

$$E[\varepsilon_{n+1,n}|\mathcal{F}_{n,n}] = 0.$$

Case 3: $n+1 \le i \le 2n-1$

Here $\varepsilon_{i+1,n} = \frac{1}{\sqrt{n_1}}\left(W_{i-n+1} - (1-W_{i-n+1})\frac{K_{n,i-n+1}}{M}\right)(Y_{i-n+1} - \mu_{W_{i-n+1}}(X_{i-n+1}))$.

Given $\mathcal{F}_{i,n}$, all W_j, X_j, $K_{n,j}$ for $j \le n$ and Y_j for $j \le i-n$ are known. The argument is identical to Case 2:

$$E[Y_{i-n+1} - \mu_{W_{i-n+1}}(X_{i-n+1})|\mathcal{F}_{i,n}] = E[Y_{i-n+1}|W_{i-n+1}, X_{i-n+1}] - \mu_{W_{i-n+1}}(X_{i-n+1}) = 0.$$

Therefore, $E[\varepsilon_{i+1,n}|\mathcal{F}_{i,n}] = 0$ for all $i = 1, \dots, 2n-1$, confirming that $\{(X_{i,n}, \mathcal{F}_{i,n})\}$ is a martingale.

□

Problem 2. ***Time-reversed martingales.***

Let $\{X_t : t \in T\}$ be a sequence of integrable random variables adapted to a sequence of sub-sigma-algebras $\{\mathcal{F}_t : t \in T\}$, where now $\mathcal{F}_t \supseteq \mathcal{F}_{t+1}$ for each t. The sequence $\{(X_t, \mathcal{F}_t) : t \in T\}$ is called a reversed martingale if $E[X_t|\mathcal{F}_{t+1}] = X_{t+1}$ for each t. At date $t+1$, the best guess of X_t having only observed information now and into the future is the value today (i.e., X_{t+1}).

(a) *Show that for each n, the sequence $X_n, X_{n-1}, \dots, X_0$ is a martingale with respect to the filtration $\mathcal{F}_n \subseteq \mathcal{F}_{n-1} \subseteq \dots \subseteq \mathcal{F}_0$.*

(b) *Use the law of iterated expectations to show $E[X_s|\mathcal{F}_t] = X_t$ for each $s < t$. In particular, $X_t = E[X_0|\mathcal{F}_t]$.*

(c) *Suppose that $E[X_0^2] < \infty$. Show that*

$$X_0 = X_n + \sum_{t=0}^{n-1} \varepsilon_t,$$

and

$$E[X_0^2] = E[X_n^2] + \sum_{t=0}^{n-1} E[\varepsilon_t^2],$$

where $\varepsilon_t = X_t - X_{t+1}$.

(d) *Deduce that $\sum_{t\ge0} E[\varepsilon_t^2] < \infty$ and therefore that $\sum_{t\ge0} \varepsilon_t$ is a well defined random variable with finite second moment.*

(e) *Show that X_n converges in L^2 to $X := X_0 - \sum_{t=0}^{\infty} \varepsilon_t$.*

(f) *Let $\mathcal{F}_\infty = \cap_{n\ge1}\mathcal{F}_n$. Show that $X = E[X_0|\mathcal{F}_\infty]$. You may assume that X is $\mathcal{F}_\infty$-measurable.*

Solution.

(a) Define $\widetilde{X}_i = X_{n-i}$ and $\widetilde{\mathcal{F}}_i = \mathcal{F}_{n-i}$ for $i = 0, 1, \ldots, n$.

Then $\widetilde{\mathcal{F}}_0 \subseteq \widetilde{\mathcal{F}}_1 \subseteq \ldots \subseteq \widetilde{\mathcal{F}}_n$ (the filtration is increasing in the forward direction), and by the reversed martingale property:

$$E[\widetilde{X}_{i+1}|\widetilde{\mathcal{F}}_i] = E[X_{n-i-1}|\mathcal{F}_{n-i}] = X_{n-i} = \widetilde{X}_i.$$

Therefore, $\{\widetilde{X}_i, \widetilde{\mathcal{F}}_i\}_{i=0}^n$ is a (forward) martingale, which is equivalent to saying $(X_n, X_{n-1}, \ldots, X_0)$ is a martingale with respect to the filtration $(\mathcal{F}_n, \mathcal{F}_{n-1}, \ldots, \mathcal{F}_0)$.

(b) For $s < t$, by iterated expectations and the reversed martingale property:

$$\begin{aligned}
E[X_s|\mathcal{F}_t] &= E[E[X_s|\mathcal{F}_{s+1}]|\mathcal{F}_t] \\
&= E[X_{s+1}|\mathcal{F}_t] \\
&= E[E[X_{s+1}|\mathcal{F}_{s+2}]|\mathcal{F}_t] \\
&= E[X_{s+2}|\mathcal{F}_t] \\
&= \cdots \\
&= E[X_t|\mathcal{F}_t] = X_t.
\end{aligned}$$

In particular, taking $s = 0$ gives $X_t = E[X_0|\mathcal{F}_t]$.

(c) From the definition $\varepsilon_t = X_t - X_{t+1}$, telescoping yields:

$$X_0 = X_0 - X_n + X_n = \sum_{t=0}^{n-1}(X_t - X_{t+1}) + X_n = X_n + \sum_{t=0}^{n-1} \varepsilon_t.$$

For the variance decomposition, note that $E[\varepsilon_t|\mathcal{F}_{t+1}] = E[X_t - X_{t+1}|\mathcal{F}_{t+1}] = X_{t+1} - X_{t+1} = 0$, so ε_t is a martingale difference sequence with respect to $\{\mathcal{F}_t\}$ (viewed in reverse).

Since X_n is $\mathcal{F}_n$-measurable and $E[\varepsilon_t|\mathcal{F}_{t+1}] = 0$:

$$E[X_n\varepsilon_t] = E[E[X_n\varepsilon_t|\mathcal{F}_{t+1}]] = E[X_n E[\varepsilon_t|\mathcal{F}_{t+1}]] = 0$$

for all $t < n$.

Similarly, for $s < t$, ε_s and ε_t are orthogonal:

$$E[\varepsilon_s\varepsilon_t] = E[E[\varepsilon_s\varepsilon_t|\mathcal{F}_{s+1}]] = E[\varepsilon_t E[\varepsilon_s|\mathcal{F}_{s+1}]] = 0,$$

since $E[\varepsilon_s|\mathcal{F}_{s+1}] = 0$ and ε_t is $\mathcal{F}_{s+1}$-measurable (as $\mathcal{F}_t \subseteq \mathcal{F}_{s+1}$ for $t > s$).

Therefore:

$$\begin{aligned} E[X_0^2] &= E\left[\left(X_n + \sum_{t=0}^{n-1} \varepsilon_t\right)^2\right] \\ &= E[X_n^2] + 2E\left[X_n \sum_{t=0}^{n-1} \varepsilon_t\right] + E\left[\left(\sum_{t=0}^{n-1} \varepsilon_t\right)^2\right] \\ &= E[X_n^2] + 0 + \sum_{t=0}^{n-1} E[\varepsilon_t^2] \\ &= E[X_n^2] + \sum_{t=0}^{n-1} E[\varepsilon_t^2]. \end{aligned}$$

(d) From part (c), for any n:

$$\sum_{t=0}^{n-1} E[\varepsilon_t^2] = E[X_0^2] - E[X_n^2] \leq E[X_0^2] < \infty,$$

since $E[X_n^2] \geq 0$. Taking $n \to \infty$:

$$\sum_{t=0}^{\infty} E[\varepsilon_t^2] \leq E[X_0^2] < \infty.$$

This implies that $\sum_{t=0}^{\infty} \varepsilon_t$ converges in L^2 (and hence is well-defined with finite second moment), because for $m < n$:

$$E\left[\left(\sum_{t=m}^{n-1} \varepsilon_t\right)^2\right] = \sum_{t=m}^{n-1} E[\varepsilon_t^2] \to 0$$

as $m, n \to \infty$, showing that the partial sums form a Cauchy sequence in L^2.

(e) Define $X = X_0 - \sum_{t=0}^{\infty} \varepsilon_t$. Then:

$$X_n = X_0 - \sum_{t=0}^{n-1} \varepsilon_t = X + \sum_{t=n}^{\infty} \varepsilon_t.$$

The L^2 distance is:

$$E[(X_n - X)^2] = E\left[\left(\sum_{t=n}^{\infty} \varepsilon_t\right)^2\right] = \sum_{t=n}^{\infty} E[\varepsilon_t^2] \to 0$$

as $n \to \infty$, since this is the tail of a convergent series. Therefore, $X_n \to X$ in L^2.

(f) From part (b), $X_n = E[X_0|\mathcal{F}_n]$ for each n. Since $X_n \to X$ in L^2 (and hence in L^1), and X is $\mathcal{F}_\infty$-measurable by assumption, for any $A \in \mathcal{F}_\infty$:

$$E[X_0\mathbf{1}_A] = E[X\mathbf{1}_A]$$

must be shown.

Since $A \in \mathcal{F}_\infty \subseteq \mathcal{F}_n$ for all n:

$$E[X_0\mathbf{1}_A] = E[E[X_0|\mathcal{F}_n]\mathbf{1}_A] = E[X_n\mathbf{1}_A].$$

Taking $n \to \infty$ and using L^1 convergence of X_n to X:

$$E[X_0\mathbf{1}_A] = \lim_{n\to\infty} E[X_n\mathbf{1}_A] = E[X\mathbf{1}_A].$$

Since this holds for all $A \in \mathcal{F}_\infty$, the conclusion is $X = E[X_0|\mathcal{F}_\infty]$.

□

Problem 3. *Tail sigma-algebra.*
Let S be a measure-preserving map on $(\Omega, \mathcal{F})$. Let $\{X_t : t \in \mathbb{N}_0\}$ be a strictly stationary stochastic process constructed using S. Recall that a set is invariant if $S^{-1}A = A$, and that $\mathcal{I}$ is the sigma-algebra of invariant sets. Let $\mathcal{F}_n = \sigma(\{X_t : t \geq n\})$ be sigma-algebras generated by the future values of X from date n onwards. The so-called tail sigma-algebra $\mathcal{F}_\infty = \cap_n \mathcal{F}_n$ is their limit. Show that $\mathcal{I} \subseteq \mathcal{F}_\infty$.

Solution. Let $A \in \mathcal{I}$, so $S^{-1}(A) = A$ by definition of an invariant set.

Since $\{X_t\}$ is strictly stationary and constructed using the measure-preserving map S, the canonical representation is:

$$X_t(\omega) = X_0(S^t(\omega))$$

for all $t \in \mathbb{N}_0$.

Therefore, the sigma-algebra $\mathcal{F}_n$ generated by $\{X_t : t \geq n\}$ can be written as:

$$\mathcal{F}_n = \sigma(\{X_t : t \geq n\}) = \sigma(\{X_0 \circ S^t : t \geq n\}) = \sigma(\{X_0 \circ S^{n+k} : k \geq 0\}).$$

By properties of sigma-algebras and measurable functions:

$$\mathcal{F}_n = (S^n)^{-1}(\mathcal{F}_0),$$

where $(S^n)^{-1}(\mathcal{F}_0) = \{(S^n)^{-1}(B) : B \in \mathcal{F}_0\}$.

Now, since A is invariant, $S^{-1}(A) = A$. By induction, this implies:

$$S^{-n}(A) = (S^{-1})^n(A) = A$$

for all $n \geq 0$.

This can be written as:

$$A = S^{-n}(A) = \{\omega : S^n(\omega) \in A\}.$$

This means that the event A is completely determined by the trajectory of ω starting from time n onward, i.e., by $(X_n(\omega), X_{n+1}(\omega), \dots)$.

More formally, on the canonical space where $\omega = (x_0, x_1, x_2, \dots)$ and $S^n(\omega) = (x_n, x_{n+1}, \dots)$, the invariance condition $A = S^{-n}(A)$ means:

$$A = \{\omega : (X_n(\omega), X_{n+1}(\omega), \dots) \in B_n\}$$

for some measurable set B_n (which may depend on n, but the event A itself does not).

This is precisely the definition of A being measurable with respect to $\sigma(X_n, X_{n+1}, \dots) = \mathcal{F}_n$.

Since $A \in \mathcal{F}_n$ for all $n \in \mathbb{N}_0$, it follows that:

$$A \in \bigcap_{n=0}^{\infty} \mathcal{F}_n = \mathcal{F}_\infty.$$

Thus, $\mathcal{I} \subseteq \mathcal{F}_\infty$. $\square$

Problem 4. ***Martingales and uniform integrability.***

Let $X_1, X_2, \dots$ be independent random variables with mass at $1, -1, 0$ with probabilities $\frac{1}{2t}, \frac{1}{2t}, 1 - \frac{1}{t}$. Let $Y_1 = X_1$. For $t \geq 1$:

$$Y_t = \begin{cases} X_t & \text{if } Y_{t-1} = 0, \\ tY_{t-1}|X_t| & \text{if } Y_{t-1} \neq 0. \end{cases}$$

Let $\mathcal{F}_t = \sigma(\{Y_1, \dots, Y_t\})$.

(a) *Show that $\{(Y_t, \mathcal{F}_t) : t \geq 1\}$ is a martingale.*

(b) *Is $\{Y_t : t \geq 1\}$ a Markov process? Explain.*

(c) *Is $\{Y_t : t \geq 1\}$ a time-homogeneous Markov process? Explain.*

(d) *Is $\{Y_t : t \geq 1\}$ uniformly integrable? Explain.*

(e) *Does $\{Y_t : t \geq 1\}$ converge almost surely? Explain.*

Solution.

(a) The verification is that $E[Y_t|\mathcal{F}_{t-1}] = Y_{t-1}$.

Case 1: $Y_{t-1} = 0$. Then $Y_t = X_t$. Since X_t is independent of $\mathcal{F}_{t-1}$:

$$E[Y_t|Y_{t-1} = 0] = E[X_t] = 1\left(\frac{1}{2t}\right) + (-1)\left(\frac{1}{2t}\right) + 0 = 0 = Y_{t-1}.$$

Case 2: $Y_{t-1} \neq 0$. Then $Y_t = tY_{t-1}|X_t|$.

$$E[Y_t|\mathcal{F}_{t-1}] = tY_{t-1}E[|X_t|].$$

The calculation yields $E[|X_t|] = |1|\left(\frac{1}{2t}\right) + |-1|\left(\frac{1}{2t}\right) + 0 = \frac{1}{t}$.

$$E[Y_t|\mathcal{F}_{t-1}] = tY_{t-1}\left(\frac{1}{t}\right) = Y_{t-1}.$$

Thus, $\{(Y_t, \mathcal{F}_t)\}$ is a martingale.

(b) Yes, $\{Y_t\}$ is a Markov process. The definition of Y_t depends only on the current state Y_{t-1} and the independent innovation X_t. The conditional probability $P(Y_t \in A|\mathcal{F}_{t-1})$ reduces to $P(Y_t \in A|Y_{t-1})$.

(c) No, it is not time-homogeneous. The transition mechanism depends explicitly on t in two ways: 1. The distribution of the innovation X_t changes with t (probability of non-zero values decreases). 2. The functional form $Y_t = tY_{t-1}|X_t|$ includes t as a multiplier.

(d) No, it is not uniformly integrable. The L^1 norm is examined. Note that $Y_t \neq 0$ if and only if $X_t \neq 0$.

$$E[|Y_t|] = P(Y_{t-1} = 0)E[|X_t|] + E[|Y_{t-1}|\mathbb{I}(Y_{t-1} \neq 0)t|X_t|].$$

Using independence:

$$E[|Y_t|] = P(Y_{t-1} = 0)\frac{1}{t} + t\left(\frac{1}{t}\right)E[|Y_{t-1}|].$$

This implies $E[|Y_t|] > E[|Y_{t-1}|]$. Specifically, approximating $P(Y_{t-1} = 0) \approx 1$, the result is $E[|Y_t|] \approx E[|Y_{t-1}|]+1/t$. This grows as the harmonic series, so $\lim_{t\to\infty} E[|Y_t|] = \infty$. An unbounded sequence in L^1 cannot be uniformly integrable.

(e) No, it does not converge almost surely. Recall $Y_t \neq 0 \iff X_t \neq 0$. $P(X_t \neq 0) = 1/t$. Since $\sum 1/t = \infty$ and X_t are independent, by Borel-Cantelli, $X_t \neq 0$ infinitely often, so $Y_t \neq 0$ infinitely often. When $Y_t \neq 0$, $|Y_t| \geq 1$ (as it takes integer values). However, $P(X_t = 0) \to 1$, so the sequence also hits 0 infinitely often. A sequence that oscillates between 0 and values with magnitude ≥ 1 infinitely often does not converge.

$\square$